

Derived Pro-Cokernel Recovery of the Loeb-Rosser Phantom

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Abstract

Pass 68 explains why the all-prime phantom is invisible at every fixed finite conductor but nonzero after applying the derived inverse limit. The quotient

$$\widehat{\mathbb{Z}}/\mathbb{Z}$$

is recovered as a derived pro-cokernel.

Main Result

Let

$$N_n = \text{lcm}(1, \dots, n).$$

At each level there is an exact sequence

$$0 \rightarrow N_n \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/N_n \mathbb{Z} \rightarrow 0.$$

The finite map $\mathbb{Z} \rightarrow \mathbb{Z}/N_n \mathbb{Z}$ is surjective, and CRT identifies

$$\mathbb{Z}/N_n \mathbb{Z} \cong \prod_{p|N_n} \mathbb{Z}/p^{v_p(N_n)} \mathbb{Z}.$$

So the finite quotient by the diagonal is zero at every level.

The kernel tower $K_n = N_n \mathbb{Z}$ is non-Mittag-Leffler, with

$$\lim_{\leftarrow} K_n = 0.$$

Applying the derived inverse limit gives

$$0 \rightarrow \mathbb{Z} \rightarrow \lim_{\leftarrow n} \mathbb{Z}/N_n \mathbb{Z} \rightarrow \lim_{\leftarrow}^1 K_n \rightarrow 0.$$

Since $\lim_{\leftarrow n} \mathbb{Z}/N_n \mathbb{Z} = \widehat{\mathbb{Z}}$,

$$\lim_{\leftarrow}^1 (N_n \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}.$$

Verification

The checker `code/scripts/check-pass68.py` generated `artifacts/reports/pass68-derived-pro-cokernel-phantom-check.json`.

Verified facts:

- the lcm tower is cofinal for moduli $1, \dots, 24$;
- CRT makes the finite cokernel zero at sampled levels;
- the kernel tower has unbounded image indices and is non-Mittag-Leffler;
- the kernel inverse limit is zero;
- completion prefixes grow while the finite cokernel remains zero.

Remaining Gap

Pass 69 should identify this derived pro-cokernel class with the earlier Loeb-Rosser boundary ϵ from the bi-complex/recollement construction, including the signed duality law $D(\epsilon) = -\epsilon^\vee$.